

SmartCity Platform Report

AI-Powered Civic Grievance System & Security Architecture

1. Project Vision & Mission

Modern municipal administrations handle thousands of civic complaints daily. Traditional grievance portals rely on manual ticket routing, which is slow, error-prone, and lacks transparent timeline forecasting. SmartCity was developed to solve this bottleneck by combining cutting-edge Artificial Intelligence with enterprise-grade privacy engineering. It provides a fast, secure, and fully automated civic ticketing pipeline that categorizes issues, forecasts resolution timelines, and notifies citizens automatically on every milestone.

2. Advanced Machine Learning (Local ML Fallbacks)

SmartCity does not rely on hardcoded rules. The platform deploys locally trained Scikit-Learn models that act as high-speed components and reliable fallbacks if the external Gemini API is rate-limited:

- * **Disposable Email Detection (Registration Spam Blocker):** Uses a Logistic Regression model with a character-level N-gram Count Vectorizer. It mathematically scans the character sequences of domains during citizen registration to block disposable domains (like @10minutemail, @tempmail), preventing bot creation.

- * **Resolution Time Forecasting:** Employs a Scikit-Learn RandomForestRegressor. It processes priority levels, departmental assignments, and seasonal parameters to calculate exactly how many days a complaint will take to resolve based on historical resolution trends, displaying a realistic timeline to the citizen.

- * **Text-Based Category Classification Fallback:** Uses TF-IDF vectorization paired with a Logistic Regression classifier to automatically parse descriptions and route tickets to appropriate municipal divisions (Water, Roads, Electricity, etc.) if Gemini is unavailable.

3. Google Gemini AI Integration Features

SmartCity integrates the Google GenAI SDK powered by the highly optimized gemini-2.5-flash model to deliver interactive, real-time civic triage and conversational features:

- * **Conversational Chatbot Assistant:** A floating 3D Bot in the citizen frontend communicates with a FastAPI route. It parses history and prompts Gemini using a tailored municipal knowledge base, allowing citizens to query reporting steps using natural language processing.

- * **Vision-Based Complaint Triage:** When citizens submit a complaint, Gemini processes both text descriptions and uploaded photos (converted to base64). It automatically detects the correct category, urgency levels, and outputs a one-sentence summary.

- * **Dynamic Category Styling:** If an admin creates a custom category, Gemini automatically generates a corresponding emoji icon and a matching hexadecimal UI color code to style the admin dashboard on the fly.

- * **AI Engineer Investigation Reports:** In the admin details panel, clicking 'View Details' triggers Gemini to instantly compile a comprehensive engineering investigation report containing impact analysis, step-by-step action plans, required equipment/personnel, and prevention recommendations.

4. Live Admin Dashboard Statistics & Analytics

The Admin Dashboard aggregates municipal data dynamically from the SQL database using optimized aggregations. Below are the current live metrics compiled for this system:

Total Tickets: 500	Open Issues: 199	Resolved: 301	Avg Resolution: 11.3 Days
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Category Distribution

Road: 139 complaints
Waste: 101 complaints
Water: 100 complaints
Electricity: 68 complaints
Drainage: 41 complaints

Priority Distribution

Medium Priority: 152 tickets
Low Priority: 151 tickets
High Priority: 120 tickets
Critical Priority: 77 tickets

5. Authentication, Validation & Automated Notifications

- * Evaluator Test Credentials:** Evaluators can use Email: 'syedareebali795@gmail.com' and Password: '123456' to login, submit complaints, and track details from the citizen portal.
- * Strict Input Validations:** The portal uses robust input validation controls. Submissions require properly formatted emails, minimum password lengths (6+ characters), and non-empty text descriptions to maintain clean databases.
- * SlowAPI Bot Protection:** To prevent automated script attacks, the complaint routes utilize SlowAPI. Multiple bots cannot spam the submission endpoint simultaneously, protecting backend resources from denial-of-service.
- * Resend Notifications (Free Tier vs Pro):** The Resend API sends automatic HTML status updates (Accept, Resolve, Reject, Escalate). In free sandbox mode, testing is strictly restricted to the verified account owner: syedareebali795@gmail.com. On a Pro/paid production plan, it will seamlessly scale to deliver alerts to all citizen emails automatically.
- * Phone Authentication Roadmap:** The phone number field is collected. Verification SMS is set as a future roadmap feature, as Firebase Phone Authentication requires a paid/pro plan for API verification.

6. Tech Stack & Deployment Matrix

Layer	Technologies	Deployment / Hosting
Frontend (Citizen & Admin)	Next.js, Tailwind CSS, Recharts, Framer Motion	Vercel
Backend API	FastAPI (Python), SQLAlchemy, SlowAPI	Railway
Database	MySQL, SQLite fallback	Railway MySQL
Authentication & Email	Firebase Auth, JWT (Admin), Resend API	Firebase / Resend